

Assignment 2: Literature Mining using ChatGPT API

Step 1: Extracting the First Abstract

The screenshot shows a PDF viewer window with the title bar 'Survey Papers' and the document title 'Multi-Scale DCNN with Dynamic Weight and Part Loss for Skin Lesion Segmentation'. The authors listed are Gaoshuai Wang*, Linrunjia Liu, Fabrice Lauri, and Amir HAJJAM El Hassani. The abstract text is highlighted in blue. A context menu is open over the abstract, showing options like 'Copy', 'Zoom In', 'Zoom Out', 'Automatically Resize', 'Zoom to Page Width', 'Zoom to Page Height', 'Split Horizontally', 'Split Vertically', 'Next Page', and 'Previous Page'.

Gaoshuai Wang*, Linrunjia Liu, Fabrice Lauri, and Amir HAJJAM El Hassani

Abstract: Accurately diagnosing skin lesion disease is a challenging task. Although present methods often use the multi-branch structure to get more clues, the rigid methods of cropping zone and fusion fail to handle the instability of the disease zone and the difference in branch results, which leads to the cropping and degrades Deep Convolutional Neural Networks (DCNN)'s performance. To solve these problems, we propose a Multi-scale DCNN with Dynamic weight and Part cross-entropy loss (MDP-DCNN) to bootstrap skin lesion diagnosis. Inspired by the object detection method, the proposed structure adjusts the cropping position based on the Gradient-weighted Class Activation Map (Grad-CAM) center. It enables the model to adapt to the disease zone variety in position and size. The proposed structure alleviates the negative influence of branch differences by comparing the grey-crop CAM. Moreover, we also propose the part cross-entropy loss to deal with the over-fitting problem. It optimizes the non-targeted label to decrease the influence on other labels' stability when the model is wrong. We conduct our model on the ISIC-2017 and ISIC-2018 datasets. Experiments demonstrate that the proposed MDP-DCNN achieves excellent results in skin lesion classification without external data. Multi-scale dynamic weight and part loss function verifies its advantages in enhancing diagnosis accuracy.

Step 2: Extracting the Second Abstract

The screenshot shows a PDF viewer window with the title bar 'Survey Papers' and the document title 'Skin Lesion Segmentation'. The corresponding author is Eliezer Soares Flores (eliezerflores@unipampa.edu.br). The work was supported in part by the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), Brazil; in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Brazil – Finance Code 001; and in part by CAPES Transformative Agreement. The abstract text is highlighted in blue. A context menu is open over the abstract, showing options like 'Copy', 'Zoom In', 'Zoom Out', 'Automatically Resize', 'Zoom to Page Width', 'Zoom to Page Height', 'Split Horizontally', 'Split Vertically', 'Next Page', and 'Previous Page'.

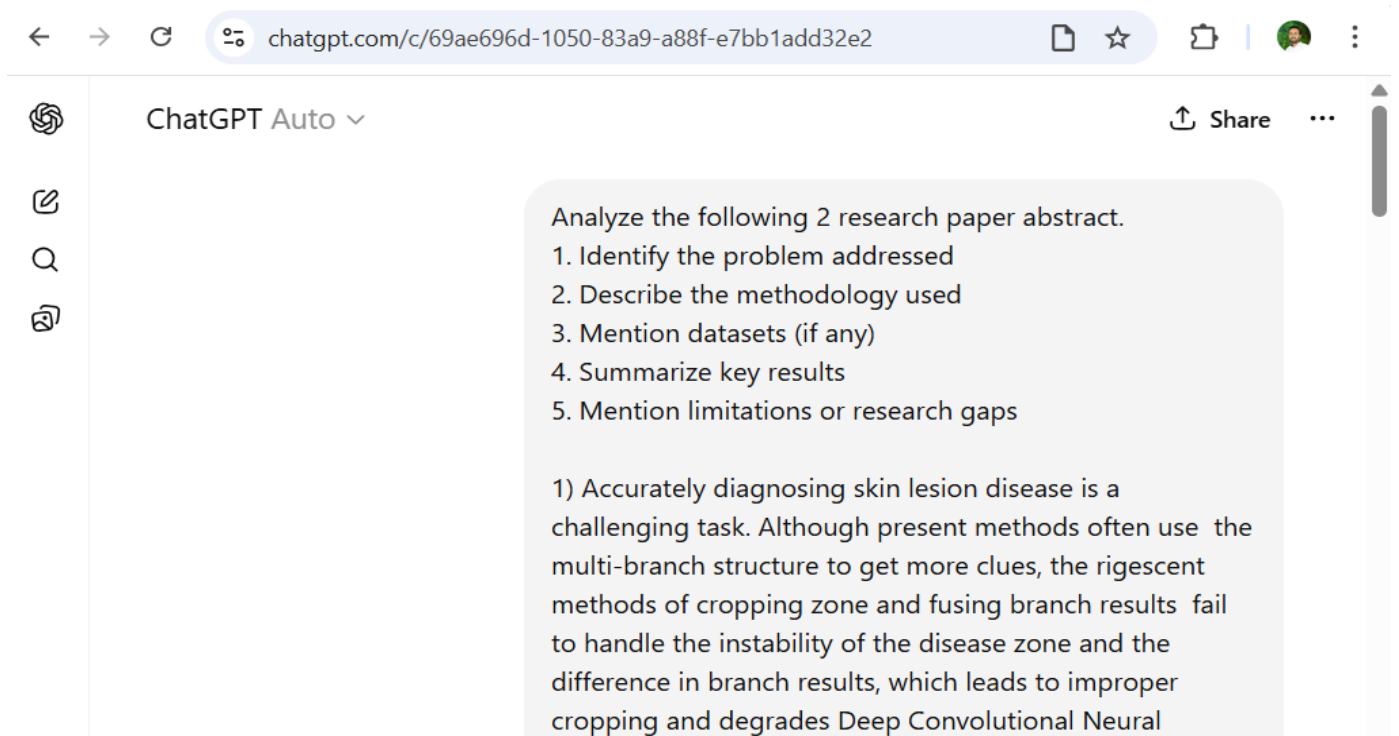
Corresponding author: Eliezer Soares Flores (eliezerflores@unipampa.edu.br)

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ABSTRACT Early detection of cutaneous melanoma is crucial for improving patient survival rates. Skin lesion segmentation is a critical step towards computer-aided diagnosis that is effective in melanoma detection. This work proposes a novel patch-based model that leverages a combination of non-negative matrix factorization and mutual information for segmenting cutaneous melanomas in clinical images. The proposed approach tends to lead to accurate skin lesion segmentations, and it was comprehensively evaluated in terms of its effectiveness based on three benchmark datasets (DermIS, DermNet, and DermQuest) using statistical analysis, ablation studies, and cross-dataset testing. The proposed method tends to improve on the segmentation results compared to published deep learning baselines (FCNet, FCN, PSPNet, SegNet, U-Net) and by recent methods specifically designed for skin lesion segmentation (DSNet, FrCNet, and LGPNet).

INDEX TERMS matrix decomposition, skin cancer.

Step 3: Structuring the AI Prompt



The screenshot shows the ChatGPT web interface. The browser address bar displays the URL: chatgpt.com/c/69ae696d-1050-83a9-a88f-e7bb1add32e2. The ChatGPT header includes the logo, 'ChatGPT Auto', a 'Share' button, and a menu icon. On the left sidebar, there are icons for editing, search, and a document. The main chat area contains a prompt in a light gray box:

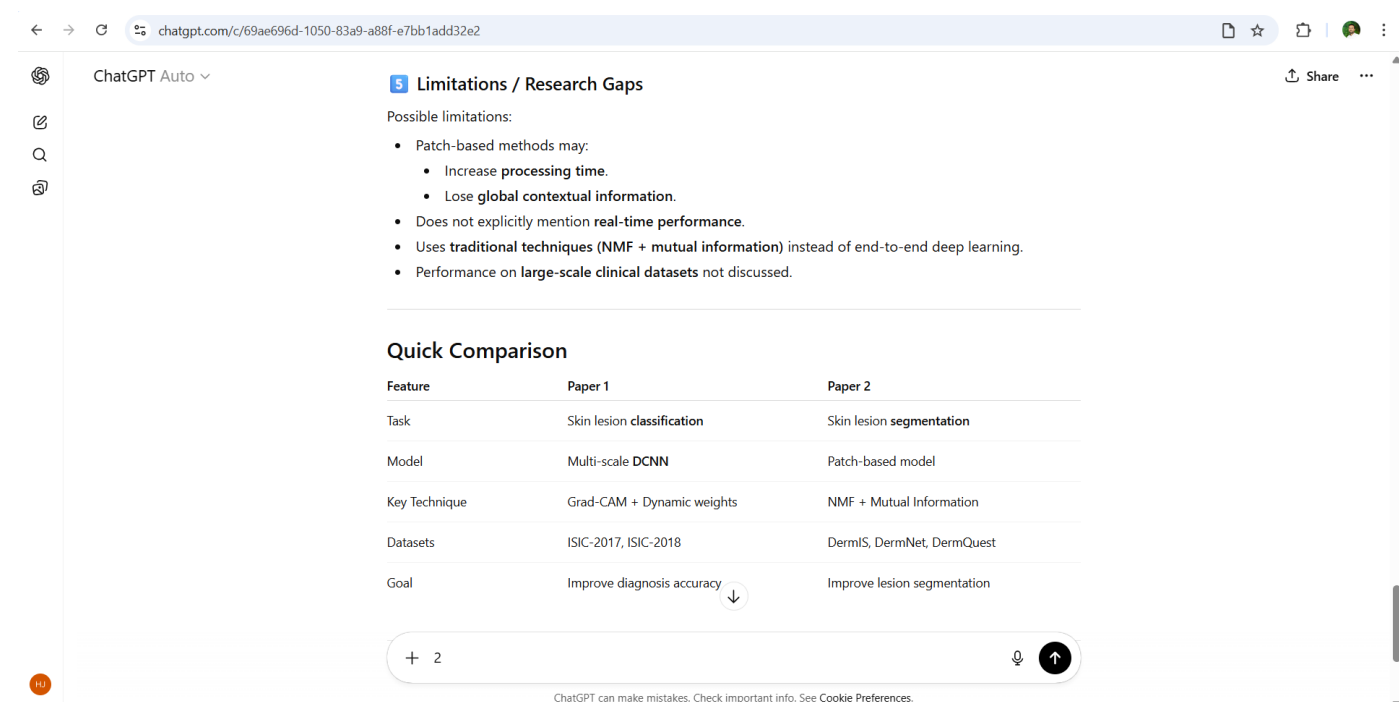
Analyze the following 2 research paper abstract.

1. Identify the problem addressed
2. Describe the methodology used
3. Mention datasets (if any)
4. Summarize key results
5. Mention limitations or research gaps

Below the list, the first abstract is provided:

1) Accurately diagnosing skin lesion disease is a challenging task. Although present methods often use the multi-branch structure to get more clues, the resescent methods of cropping zone and fusing branch results fail to handle the instability of the disease zone and the difference in branch results, which leads to improper cropping and degrades Deep Convolutional Neural

Step 4: Generating the Comparison and Analysis



The screenshot shows the ChatGPT web interface with the generated analysis. The browser address bar displays the URL: chatgpt.com/c/69ae696d-1050-83a9-a88f-e7bb1add32e2. The ChatGPT header includes the logo, 'ChatGPT Auto', a 'Share' button, and a menu icon. On the left sidebar, there are icons for editing, search, and a document. The main chat area displays the following analysis:

Limitations / Research Gaps

Possible limitations:

- Patch-based methods may:
 - Increase processing time.
 - Lose global contextual information.
- Does not explicitly mention real-time performance.
- Uses traditional techniques (NMF + mutual information) instead of end-to-end deep learning.
- Performance on large-scale clinical datasets not discussed.

Quick Comparison

Feature	Paper 1	Paper 2
Task	Skin lesion classification	Skin lesion segmentation
Model	Multi-scale DCNN	Patch-based model
Key Technique	Grad-CAM + Dynamic weights	NMF + Mutual Information
Datasets	ISIC-2017, ISIC-2018	DermIS, DermNet, DermQuest
Goal	Improve diagnosis accuracy	Improve lesion segmentation

At the bottom of the chat area, there is a text input field with a plus icon, a '2' character, a microphone icon, and an upward arrow icon. Below the input field, a small disclaimer reads: 'ChatGPT can make mistakes. Check important info. See [Cookie Preferences](#)'.